Md Obayed Reza Hridoy

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Electrical and Electronic Engineering student at BRAC University, also pursuing an associate degree in Computer Science at the University of the People. Interested in embedded systems, firmware development, embedded AI, and TinyML, with a focus on bringing deep learning and computer vision to resource-constrained hardware. Research experience in computer vision and image-based detection at LaSSET; co-leads the AI & Vision sub-team at BRACU Alter; co-founder of Tornetec and GreenSync.

Education

BRAC University — Dhaka, Bangladesh
Bachelor of Science in Electrical and Electronic Engineering — *2025 – Present*

University of the People — Online
Associate Degree in Computer Science — *2026 – Present*

Experience

Co-Founder, Head of Research & Management — 2022 – Present
Tornetec — *Dhaka, Bangladesh*

- Leading research, strategic planning, and team management for an IT startup focused on AI integration into everyday human life.
- Overseeing prototype development and ensuring alignment with real-world technology needs.

Undergraduate Researcher — 2025 – 2026
Laboratory of Space System Engineering Technology (LaSSET), BRAC University — *Dhaka, Bangladesh*

- Working on computer vision and image-based detection tasks involving deep learning and CNN-based approaches.
- Developing and evaluating detection-based solutions for research applications using Python, MATLAB, Simulink, and PyTorch.

Founder & Manager — 2022 – 2024
Zyphrst

- Authorized NFT Selling Studio at Rarible Marketplace.
- Conducting in-depth market research to identify trends and opportunities in the NFT space.

Projects

GreenSync | *Co-Founder, Systems Architect & Lead Researcher* — 2025 – Present

- Leading project planning, research coordination, and development of a technology-driven smart waste management system utilizing a reward-based user engagement model.
- Serving as the founding Project Manager and Co-Founder while contributing to ongoing research activities.
- University Innovation Hub Program (UIHP) & Hult Prize – Grant-Awarded Project.

BRACU Alter | *Co-Lead, AI & Vision Sub Team* — 2025 – 2026

- Working on deep learning and computer vision for image-based detection in disaster environments for rover navigation.
- Training and fine-tuning models for real-time multi-class object detection on embedded vision hardware.

Achievements

Champion, Ideation 5.0

Second Runner-Up, Hult Prize – BRAC University 2025

National Round Participant, Hult Prize Bangladesh

Technical Skills

Languages: C/C++, Python, Java
Engineering Tools: MATLAB, Simulink, PSpice
AI & Computer Vision: Computer Vision, PyTorch, Deep Learning, CNN, Image/Object Detection
Research: Data Analysis and Visualization, Quantitative Research Methods, Research Planning & Execution
Professional: Project Management, Interdisciplinary Collaboration, Strategic Planning